

SABIN PANTA

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EDUCATION

Utah State University

2025 Aug - May 2027(Expected)

MSc, Intelligent Water Systems

Relevant Coursework: Hydroinformatics, Advanced Databases, Deep Learning, Cloud Computing (AWS), Physical Hydrology, Hydraulic Modelling

MS Thesis: Enabling Adaptive Sample and Situational Awareness in Hydrologic Monitoring Using IoT Devices and MQTT Communication Protocol

Advisor: Dr Jeff Horsburgh

Institute of Engineering, Nepal

2015 - 2019

BE, Civil Engineering

Relevant Coursework: Statistics, C Programming, Hydrology

Thesis: Dam Breach Analysis Using HEC-RAS: A Case Study of Naumure Dam, Nepal

Advisor: Dr. Ram K Regmi

RESEARCH EXPERIENCE

Utah Water Research Laboratory

Graduate Research Assistant

Aug 2025 - Present

- Wrote an Arduino C++ wrapper that maps raw sensor readings to the OGC SensorThings schema and publishes them to HydroServer over MQTT and HTTP.
- Developed HSMQTTBridge to bridge incoming MQTT message via producer consumer queue and publish them to HydroServer.
- Implemented a SQLite based retry cache with failure classification and batch resubmission to guarantee at least once delivery during HydroServer outages.
- Containerized HSMQTTBridge via Docker Compose and configured GitHub Actions CI running linting and pytest.
- Configured a CR350 datalogger using LoggerNet to publish observations to the MQTT broker.
- Deployed Mosquitto broker on Raspberry Pi with authenticated client access, ACL, and QoS 1 delivery for reliable sensor telemetry.

Department of Civil Engineering

Undergraduate Researcher

2018-2019

- Simulated dam breach analysis of a multipurpose reservoir using HEC-RAS, conducted sensitivity analysis on breach parameters and preparation of breach hydrograph and flood inundation map using RAS Mapper and ArcGIS.

INDUSTRIAL EXPERIENCE

Jankari Tech Pvt. Ltd

Software Developer.

April 2023-April 2025

- Built scalable data ingestion pipelines for cybersecurity feeds from web sources and designed database schemas for storing them using SQLAlchemy and Alembic.
- Built CI/CD pipelines with GitHub Actions and implemented Test Driven Development with Contract and Unit Tests.

Hydro Engineering and Management Service Pvt. Ltd, Nepal

Jan 2022 - April 2023

Civil Engineer

- Performed hydrological data analysis and visualization using Python.

- Implemented Python scripting within ArcGIS to automate geospatial data processing, hydrological modeling, and spatial analyses of watersheds and flood-prone areas.

Kalika Construction Pvt Ltd, Nepal

Feb 2021 - Aug 2021

Office Engineer

- Managed bill of quantities and drafted civil drawings.

RamSagar Builders, Nepal

Civil Engineer.

Mar 2020 - Jan 2021

- Prepared cost estimates government infrastructure projects

Cloud Factory, Nepal

April 2019 - Aug 2019

Data Specialist

- Data analysis and classification, data annotation and processing for AI systems

SKILLS

Languages: Python, C, C++, JavaScript, PHP

Databases: MSSQL, SQLite

DevOps & Cloud: Docker, GitHub Actions, Drone CI, Linux, AWS EC2, AWS Lambda

Testing & QA: Playwright, Selenium, Pytest

TRAINING AND CERTIFICATION

NMA DeepLearning Summer School-2022

128 hours International Virtual Summer School Organized by NeuroMatch Academy on Deep Learning, Neural Network, RNN, LSTM and Transformer.

<https://portal.neuromatchacademy.org/certificate/d485d304-f91d-4721-837f-184c29d2b806>

PROFESSIONAL AFFILIATIONS

Nepal Engineering Council

Licensed Engineer

AWARDS AND SCHOLARSHIP

Government of Nepal Engineering Scholarship

2015

Awarded full undergraduate tuition funding for ranking 105th of 15,000 candidates in the national engineering entrance examination.